

Scalar implicatures (=SIs) as metalinguistic negation of scalar alternatives

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Introduction

the set of scalar alternatives [or, and]

- (1) a. John is laughing or crying.
b. John is both laughing and crying.
c. John is not both laughing and crying.

a set of scalar alternatives: [no, some, many, (most), all/every].

- (2) a. Some women cried.
b. All women cried.
c. Not all women cried.
- (3) John did not read many books.

Various scales

- (4) Other Examples of Horn scales:
- a. Negative quantifiers: [*not all* < *few* < *none*]
 - b. Predicates: [*cute* < *beautiful* < *stupendous*],
[*discrete* < *good* < *excellent*], ...
 - c. Numerals: [1 <... n <....]
 - d. the modals: [*possibly* < *necessarily*], [*may* < *must*],
...

Disjunction

- (5) a. (Right now) Mary is either working on her paper or seeing some of her students.
- b. Mary is not both working on her paper and seeing some of her students.
- c. Mary is neither working on her paper nor seeing all of her students.
- d. Mary is working on her paper or seeing some, but not all, of her students.

the use of *or* implicates (5b)

Prediction: (5c) → canceled by the assertion

Intuitively correct SI: (5d)

Sauerland's solution

- ▶ For every stronger alternative S' for a sentence S , a **primary SI** ('don't-know' type) is derived.
- ▶ For each primary SI, the **secondary SI** (of 'know-not' type) is checked against the primary SIs plus the assertion.
- ▶ An alternative set of ' S_1 or S_2 ' is assumed to be $\{S_1$ and S_2, S_1, S_2, S_1 or $S_2\}$. (Instead of $\{\text{and, or}\}$)

Example

- (6) Kai had the broccoli or some of the peas.
 $\leadsto$ Kai didn't have the broccoli and some of the peas.
 $\leadsto$ Kai didn't have all of the peas.
- (7) a. $\neg K(\text{Kai had some of the peas})$
 b. $\neg K(\text{Kai had the broccoli})$
 c. $\neg K(\text{Kai had the broccoli and some of the peas})$
 d. $\neg K(\text{Kai had all of the peas})$
 e. $\neg K(\text{Kai had the broccoli or all of the peas})$
 f. $\neg K(\text{Kai had the broccoli and all of the peas})$

Problem with Sauerland's solution

There are cases where Sauerland's SIs are still overly strong.

- (8) Kai had the broccoli or some of the peas last night. If he had the broccoli, he had more/all of the peas.

Sauerland's predicted SI: Kai did not have all of the peas. = (A)
(A) + the conditional = Kai didn't have the broccoli.
Intended meaning of (8): Kai had the broccoli or some, but not all, of the peas.

SIs in existential quantifiers (I)

The Neo-Gricean analysis predicts too strong SIs:

- (9)
- a. Some students who read **some** of the stories will get a good grade.
 - b. NOT(all students who read some of the stories will get a good grade)
 - c. No student who read all the stories will get a good grade. (predicted)
 - d. Some students who did not read all of the stories will get a good grade. (preferred)

SIs in existential quantifiers (II)

- (10) a. Some students made a presentation or wrote a term paper.
- b. Some students did not (both) make a presentation and write a term paper.
- c. Some students made a presentation or wrote a term paper, but did not do both.

(10a) can generate the SI in (10b).

Problem: the students who made a presentation or wrote a term paper are not guaranteed to be the same group as the students in (10b).

a more plausible interpretation of (10a): (10c)

SIs in belief contexts

– A globalistic account leads to a weak implicature.

- (11) a. John believes that some students are waiting for him.
b. John does not believe that all students are waiting for him.
c. John believes that not all students are waiting for him.

(11b) means that John has no belief at all about whether all students are waiting for him. This is too weak;

(11c) is the correct one.

Chierchia's (2004) localistic approach (I)

direct SIs

- ▶ Calculation of a SI applies to LFs.

- (12) a. Mary is either working at her paper or seeing some of her students.
- b. $[_{IP} \text{ Mary}_i [_{VP} t_i \text{ is working at her paper }] \text{ or } [_{VP} \text{ some of her students}_j [_{VP} t_j \text{ seeing } t_j]]]]$

- ▶ SIs are calculated locally.
(A SI is calculated when the VP where *some*-phrase is adjoined is interpreted.)

Chierchia's localistic approach (II)

- ▶ If compositional interpretation runs into a DE operator, a SI is blocked.

- (13) a. If Paul or Bill come, Mary will be upset.
b. If Paul comes, Mary or Sue will be upset.

- ▶ **indirect SIs** are calculated at the point where a DE operator is introduced.

- (14) I doubt John drinks and drives.

I NOT(doubt John drinks or drives) (doubt = not believe)
= I believe John drinks or drives.

Correct predictions

- (5) a. (Right now) Mary is either working at her paper or seeing some of her students.
b. Mary is either working at her paper or seeing some (though not all) of her students.
- (9) a. Some students who read some of the stories will get a good grade.
b. Some students who read some of the stories, but not all, will get a good grade.
- (11) John believes that some students are waiting for him.
 $\rightsquigarrow$ John believes that not all students are waiting for him.

Problem (I) – Indirect SIs locally calculated

- (15) a. John believes that some students are waiting for him.
- b. John does not doubt that some students are waiting for him.
- (16) a. Some students who watched TV or played games failed maths.
- b. It is not the case that no students who watched TV or played games failed maths.
- c. NOT[It is not the case that no students who watched TV and played games failed maths]
 (= No students who watched TV and played games failed maths.)

(15a) (15b) but the SI for the former is a direct one while the

Problem (II) – Global SI are necessary

- Direct SIs may be required to be globally calculated.
- (17) Every student at MIT has read *LGB* or *Syntactic Structures*. (Sauerland 2004, (58))
 - ↪ NOT[Every student at MIT has read *LGB* and *Syntactic Structures*]
 - ↪ No student at MIT has read *LGB* and *Syntactic Structures*.
- (18) Everyone must solve some of the problems.
 - ↪ NOT[everyone must solve all of the problems]
 - ↪ Everyone must not solve all of the problems.

Problem (II) -continued

- (19) It's possible that Paul ate some of the eggs. (Hirschberg 1985, (75a))
 $\leadsto$ It's not possible that Paul ate all of the eggs.
 $\nrightarrow$ It is possible that Paul did not eat all of the eggs.
- (20) Some students who drank some of the beer were allowed to drive.
 $\leadsto$ NOT(some students who drank all of the beer were allowed to drive)
 $\nrightarrow$ Some students who drank some, but not all, of the beer were allowed to drive.

Quantifier raising clause-bound?

–Why QRs are bounded clausally

- (21) Some students_i [John believes that [t_i are waiting for him]]

Chierchia probably needs to assume a certain constraint on QRs of scalar terms to get what he wants to get.

SIs as metalinguistic negation (I)

– Disjunction

- (22) a. John walked or read some of the short stories.
b. $ALT = \{ \text{John walked or read some of the ss,}$
 $\text{John walked or read many of the ss,}$
 $\text{John walked or read all of the ss} \}$

All alternatives allow the possibility that the first disjunct holds.

Question: What kind of negation allows such a possibility?

Answer: Metalinguistic negation

Implicit metalinguistic negation

- (23) Mary: What is John doing these days?
Bill: He is still raising two mongeese.
(He is not raising two mongooses.)
- (24) A: Maggie is smart and attractive.
B: She is not smart and attractive; she is only attractive.
- (25) A: I like Sue. She's smart and attractive.
B: She's attractive.
- (26) (John walked or read some of the short stories.)
John didn't walk or read all of the short stories(
because it is not possible that he read all the short
stories).

Appropriateness condition

An information state & the meaning of a sentence: a set of possible worlds

- (27) a. An information state S verifies a proposition ϕ iff $S \cap \phi = S$.
- b. An information state S falsifies a proposition ϕ iff $S \setminus \phi = S$. (Here $\setminus$ is a set subtraction.)
- (28) A sentence ϕ is appropriate to utter in an information state S iff
- (i) ϕ is verified in S and
 - (ii) there is no $\phi' \in \text{ALT}(\phi)$ such that ϕ' is stronger than ϕ and ϕ' is verified in S .

Appropriateness conditions – continued

- (29) By the use of a scalar term σ in an utterance of α , the speaker S implicates any one in $\{\text{NOTM}[\beta] \mid \beta \neq \alpha, \beta \in \text{ALT}_\sigma(\alpha)\}$.
- (30) Preliminary SIs (=PSIs) of an utterance of α by the use of a scalar term σ are $\{\text{NOTM}[\beta] \mid \beta \neq \alpha, \beta \in \text{ALT}_\sigma(\alpha), \beta \text{ is stronger than } \alpha\}$
- (31) Suppose that α has the form of $R(A)$, where A is a constituent of semantic type t containing a scalar term σ and R is the rest of the sentence, and that β is a stronger alternative of α which has the form of $R(B)$. Then $\text{NOTM}[\beta] = \text{NOT}[B]$, where B is stronger than A .

Truth-functional operators

- (32) John walked or read some of the short stories.
- (33) a. A='John walked';
B='John read some of the short stories';
C='John read all of the short stories';
 $ALT_{or}(A \text{ or } B) = \{A \text{ and } B, A, B, A \text{ or } B\}$;
 $ALT_{some}(A \text{ or } B) = \{A \text{ or } B, A \text{ or } C\}$, where $C \subseteq B$.
- b. SIs from *some*: NOTM[A or C] =
(i) NOT[A or C], (ii) NOT[C]
- (34) i. NOT[C] & [A or B]
ii. [A or [B & NOT[C]]]

Truth-functional operator – continued

In (32), (i) is preferred, but in (8'), the most preferred reading is canceled and (ii) is adopted.

- i. NOT[C] & [A or B]
- ii. [A or [B & NOT[C]]]

(32) John walked or read some of the short stories.

(8') Kai had the broccoli or some of the peas last night. If he had the broccoli, he had more/all of the peas.

A = 'Kai had the broccoli';

B = 'Kai had some of the peas';

C = 'Kai had all of the peas';

$A \subseteq C \subset B$

Non-truth-functional operators – universal quantifier

- (35) a. Every student read some of the stories.
b. $ALT_{some}((35a)) =$
 {Every student read some of the stories,
 Every student read all of the stories}
c. NOTM[every student read all of the stories]
- (36) a. NOT[every_x student [x read all of the stories]]
b. NOT[x read all of the stories]

(36b) is a local SI, but it cannot move up to the main context because x becomes free. So we get two SIs, as given in (37)

- (37) a. Not every student read all of the stories.
b. Every student read not all of the stories.
 (= No students read all of the stories.)

Non-truth-functional operators – existential quantifiers

- (20') a. Some students who drank some of the beer were allowed to drive.
- b. NOTM[Some students who drank all of the beer were allowed to drive]
- c. i. NOT[some_x students who [x drank all of the beer were allowed to drive]]
- ii. NOT[x drank all of the beer]
- (38) a. No students who drank all the beer were allowed to drive & [some students who drank some of the beer were allowed to drive]
- b. Some students who drank some, but not all, of the beer were allowed to drive.

Non-truth-functional operators – epistemic operators

- (39) a. John believes that some of the students are waiting for him.
b. NOTM[John believes that all of the students are waiting for him]
- (40) a. NOT[John believes_w that all of the students are waiting for him in w]
b. NOT[all of the students are waiting for him in w]
c. i. NOT[all of the students are waiting for him in w] & John believes that some of the students are waiting for him.
ii. John believes that some of the students are waiting for him & that NOT[all of the students are waiting for him in w].

Informativeness

- (19') a. It's possible that Paul ate some of the eggs.
b. NOTM[It's possible that Paul ate all of the eggs]
c. i. NOT[It is possible_w that Paul ate all of the eggs
in w]
ii. NOT[Paul ate all of the eggs in w]
- (41) a. Some students who drank some of the beer are
allowed to drive.
b. No students who drank all of the beer are allowed to
drive. (global)
c. Some students who drank some, but not all, of the
beer are allowed to drive. (local)

Contextual plausibility (I)

(42) prefers the local SI, while (43) prefer the global SIs.

(42) Every student wrote a paper or made a classroom presentation. (Chierchia 2002)

(43) Every student at MIT has read *LGB* or *Syntactic Structures*.

When a student is supposed to do an assignment, each student is likely to satisfy the minimal requirement.

Students studying linguistics at MIT are likely to read as many syntax books as possible.

Contextual plausibility (II)

- (44) a. Everyone must solve some of the problems.
b. NOTM[Everyone must solve all of the problems]
c. NOT[everyone_x must_w [x solve all of the problems in w]]
d. (Everyone_x) NOT[must_w [x solve all of the problems in w]]
e. (Everyone_x must_w) NOT[x solve all of the problems in w]
- (45) a. Every student is allowed to read some of the comic books.
b. NOT[every_x student is allowed_w [x to read all of the comic books in w]]
c. (Every_x student is) NOT[allowed_w [x to read all of the comic books in w]]
d. (Every_x student is allowed_w) NOT[x to read all of the comic books in w]

No binding problem

- (46)
- a. Some students who read some of the stories will get a good grade.
 - b. NOT(some students who read all of the stories will get a good grade)
 - c. Some students who NOT(x read all of the stories) will get a good grade.

– The stronger implicature (= (46b)) is canceled for contextual implausibility.

– In (46c), the students mentioned in the implicature are guaranteed to be the same group as those mentioned in the assertion.

Conclusion

- ▶ SIs are not always calculated locally, as Chierchia (2002) claims.
- ▶ SIs are calculated from metalinguistic negation, not from descriptive negation.
- ▶ Metalinguistic negation allows multiple SIs.
- ▶ Informativeness, plausibility, etc determine which one is the actual one.
- ▶ Local SIs behave differently depending on whether they are embedded in truth-functional operators or in non-truth-functional operators.